

Morgan Patel

+1 (555) 010-4410 | morgan.patel@example.com | example.com/code/mpatel

Date of birth: 21 Jul 1998 | Nationality: Indian

EDUCATION

Ridgeway University | Ph.D. candidate in Statistics | Metro City, USA | Sep 2021 - Expected May 2026

Ridgeway University | B.S. in Mathematics | Metro City, USA | Sep 2016 - May 2020

EXPERIENCE

Ridgeway University | Graduate Research Assistant, Statistical Learning Lab | Metro City, USA | Jun 2020 - Aug 2021

- Built the lab's simulation pipeline on the shared cluster, cutting a 2,000-run Monte Carlo study from 3 days to 5 hours and making runs reproducible by seed.
- Derived a variance bound for a sparse regression estimator that tightens the previous bound by a log factor; it is now Section 3 of a paper under review at JASA.
- Taught weekly recitations for 60 students in graduate probability and writes the problem sets, earning a 4.8/5 teaching rating.
- Released an open-source R package for high-dimensional covariance estimation while maintaining the lab's shared cluster, organizing the weekly reading group and grading for two courses, which was downloaded 3,000 times in its first year.

Northpeak Capital | Quantitative Research Intern | Metro City, USA | Jun 2025 - Aug 2025

- Built a short-horizon order-book imbalance signal for liquid index futures that added 0.4 Sharpe to the desk's book over 18 months of out-of-sample backtest after costs.
- Improved risk-adjusted returns by 35% with a cost-aware position smoother on the desk's futures signal.
- Validated the signal on 6 years of tick data with purged walk-forward splits and an embargo period, so no look-ahead information leaked into training.
- Wrote a feature store for 120 microstructure features with point-in-time joins, which the research team reused in two later signal projects.
- Presented the signal, its capacity estimate and failure cases to the portfolio managers, who approved a small live allocation for the next quarter.
- Documented the backtest assumptions, transaction-cost model and known failure regimes in the desk's research wiki for future interns.

PROJECTS

Volatility Forecasting Study | Independent Research | Python, PyTorch | Jan 2024 - Present

- Beat a HAR-RV baseline's out-of-sample QLIKE loss by 7% on 30 equity indices with a temporal convolutional model trained on realized-volatility features.
- Cut forecast error from 0.20 to 0.15, a 33% improvement, by adding realized-volatility features and an asymmetric loss.
- Ran the robustness study on the VX9 backtest harness and wrote up the results as a 12-page working paper.

Kaggle Market Prediction Competition | Team of 3 | Python | Jun 2023 - May 2023

- Placed 41st of 2,900 teams with a gradient-boosting ensemble on 300 engineered features, finishing in the top 2% of the private leaderboard.
- Cut validation leakage by switching to time-grouped folds, which closed a 0.02 gap between local validation and leaderboard scores.
- Added 0.4 to the Sharpe ratio of a futures book with an order-book imbalance signal, tested over 18 months out of sample.

SKILLS

Programming: Python, R, SQL, C++, Kafka

Methods: time-series econometrics, Bayesian inference, gradient boosting, PyTorch